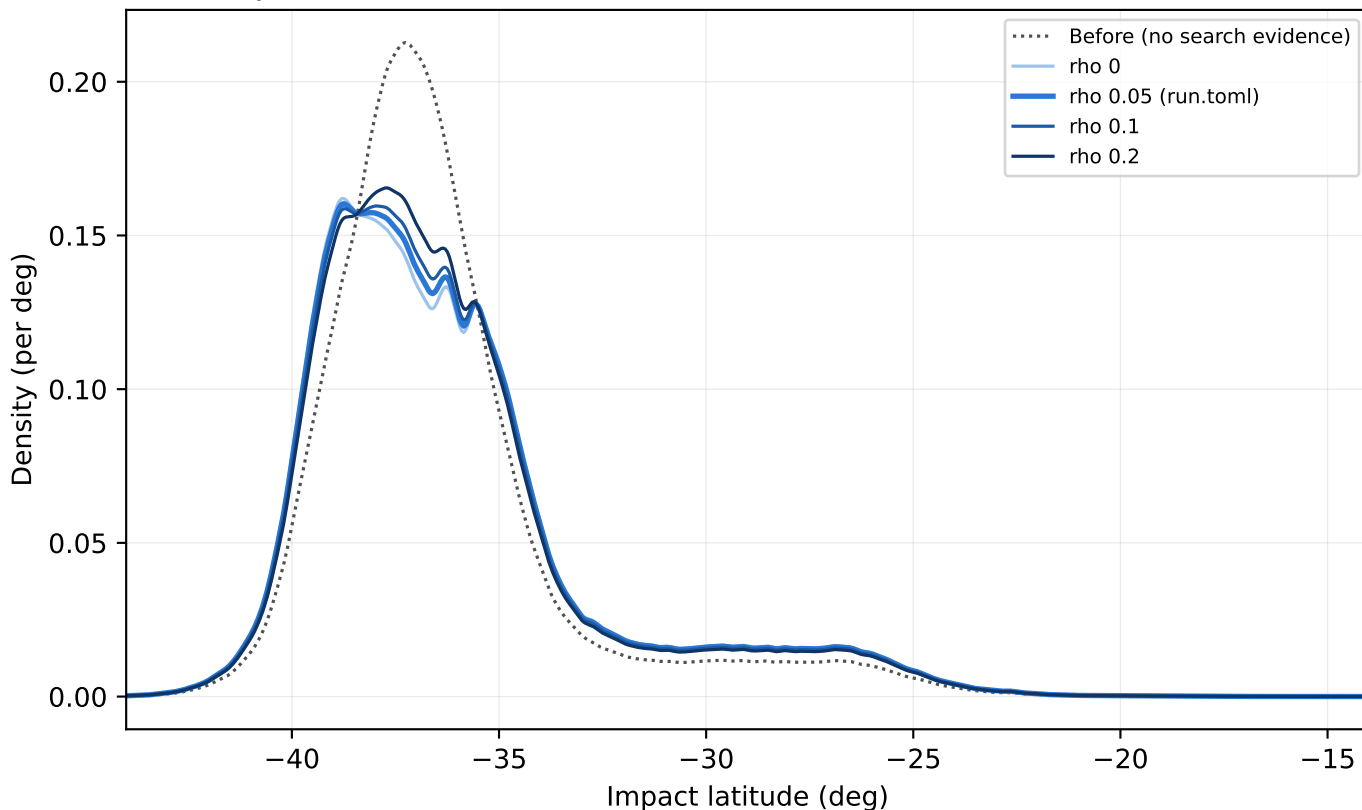
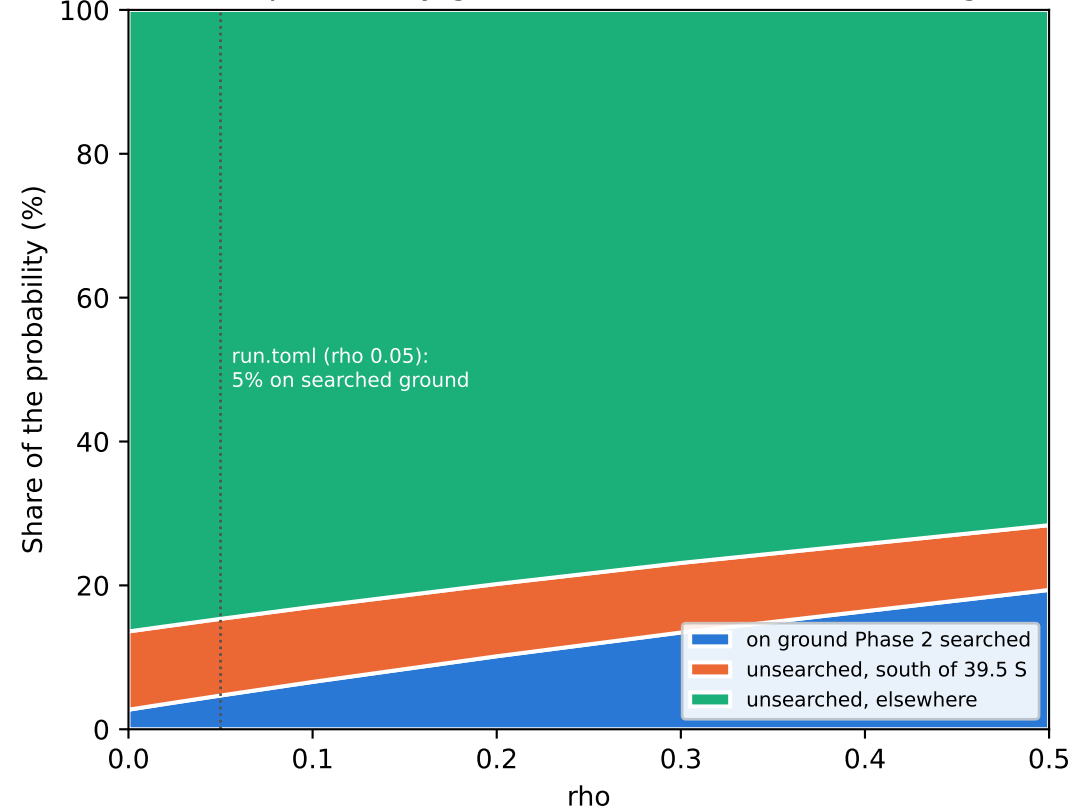


Impact latitude with the Phase 2 and Bluefin searches as evidence

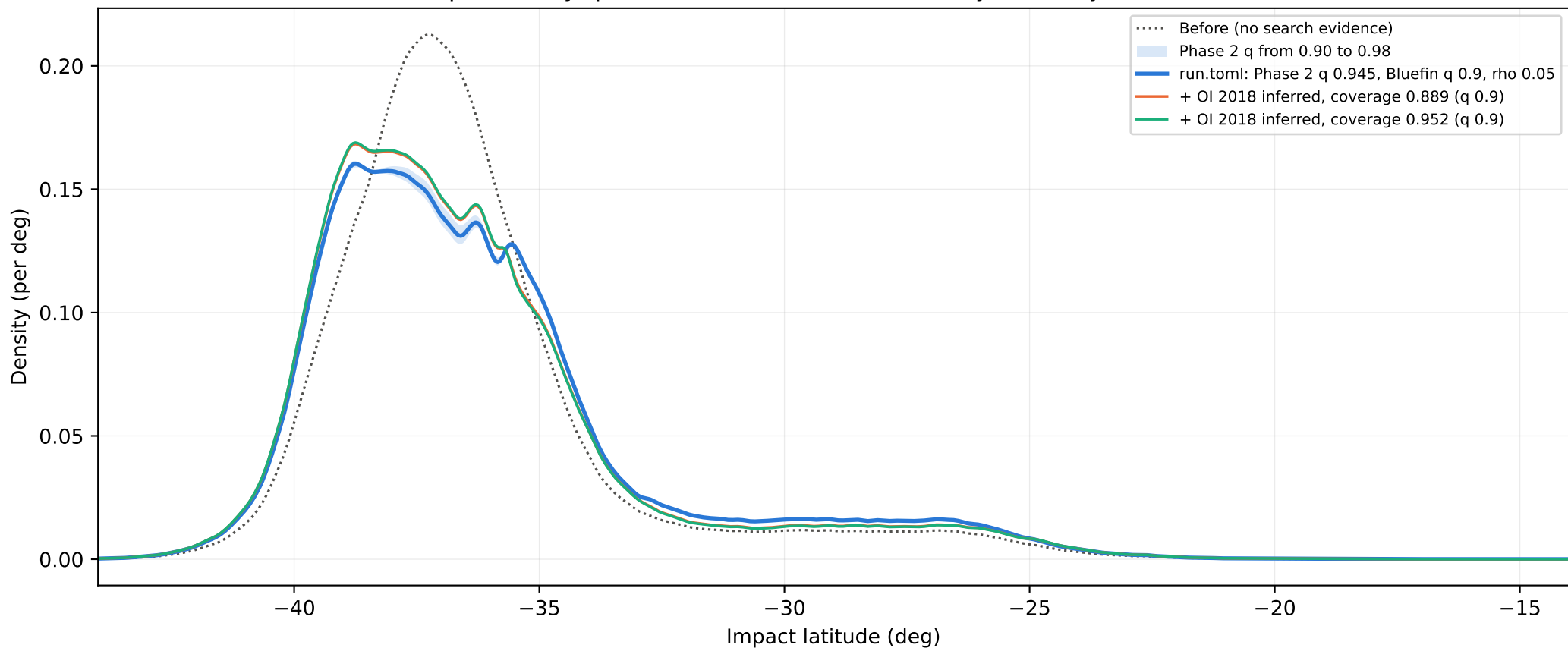


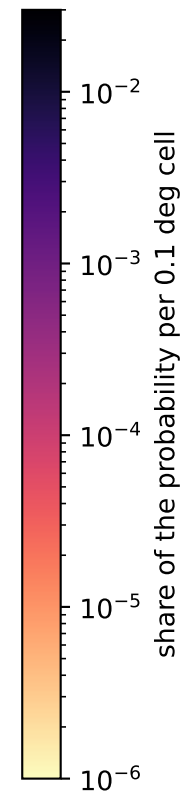
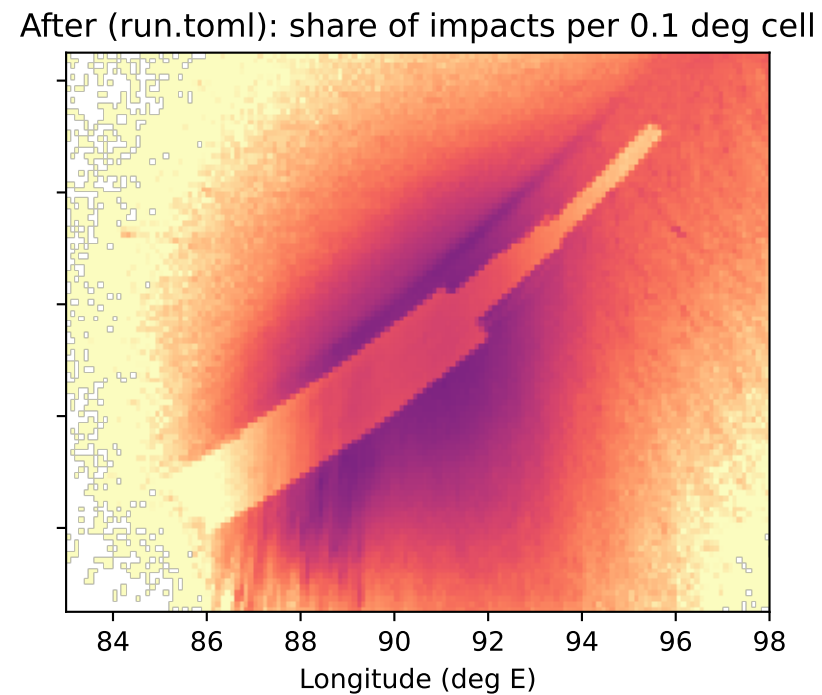
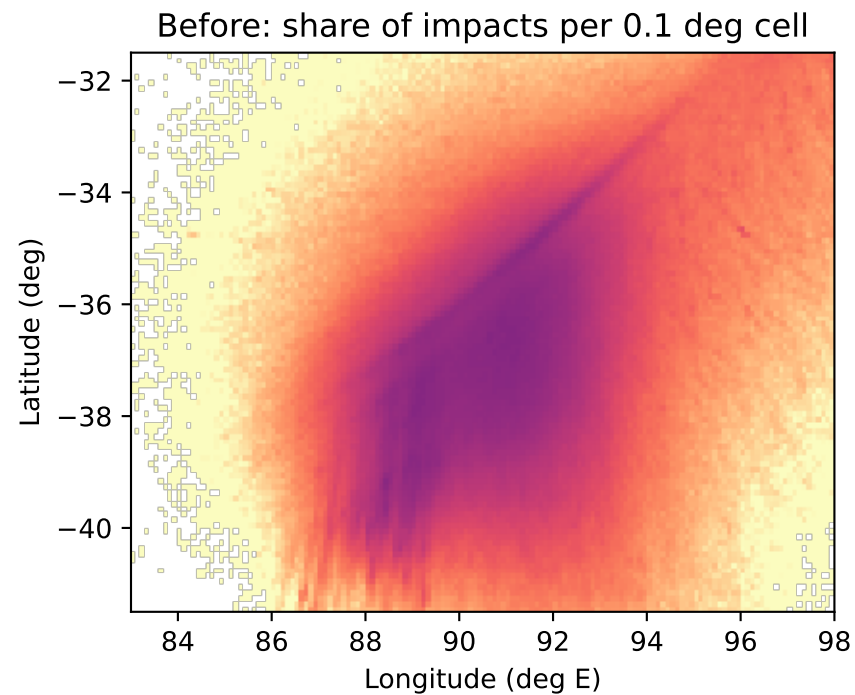
Where the probability goes (before: 31% on searched ground)



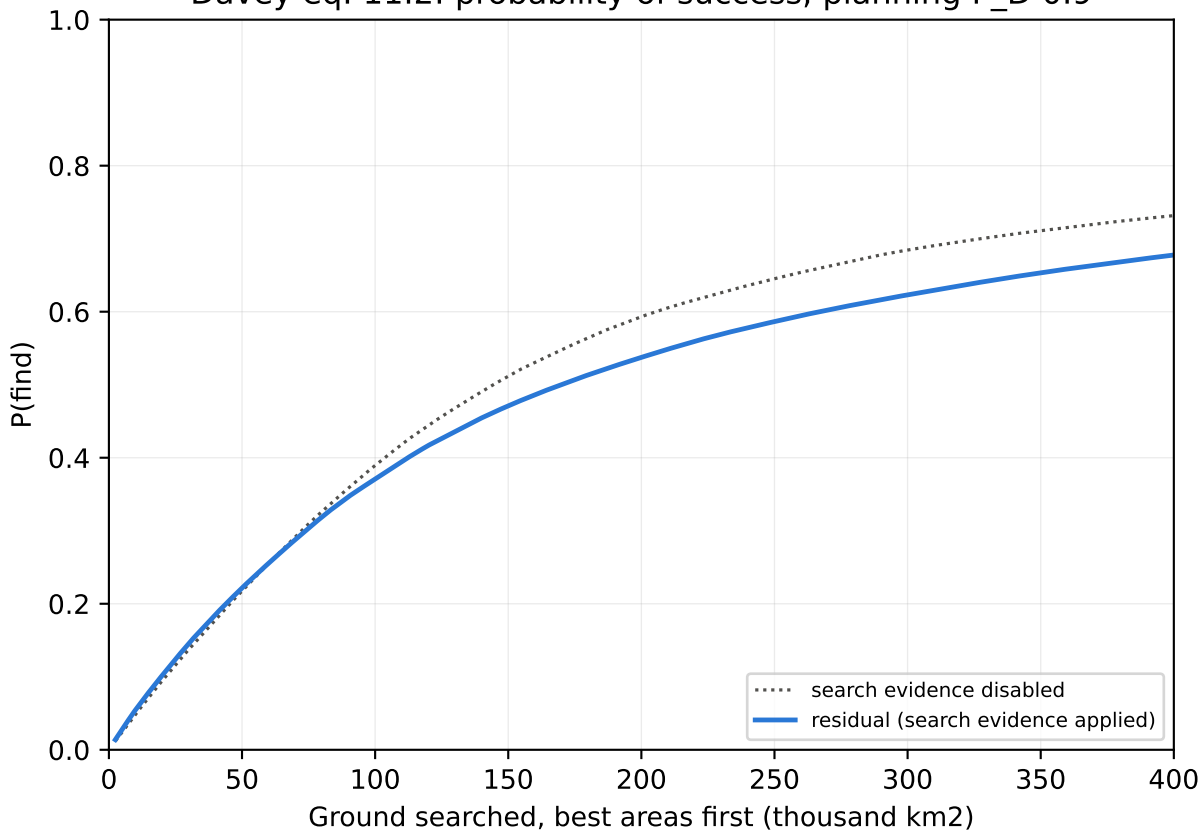
rho is the chance that the wreck could not have been found even where the sonar looked: hidden by terrain, buried, or imaged but dismissed. ATSB's detection ratings (q) cover data quality; rho is what they do not. At rho 0, 3% stays on searched ground; 68.9% of the impacts before the search lie off it.

Detection probability q , and the inferred Ocean Infinity 2018 layer (labelled variant)

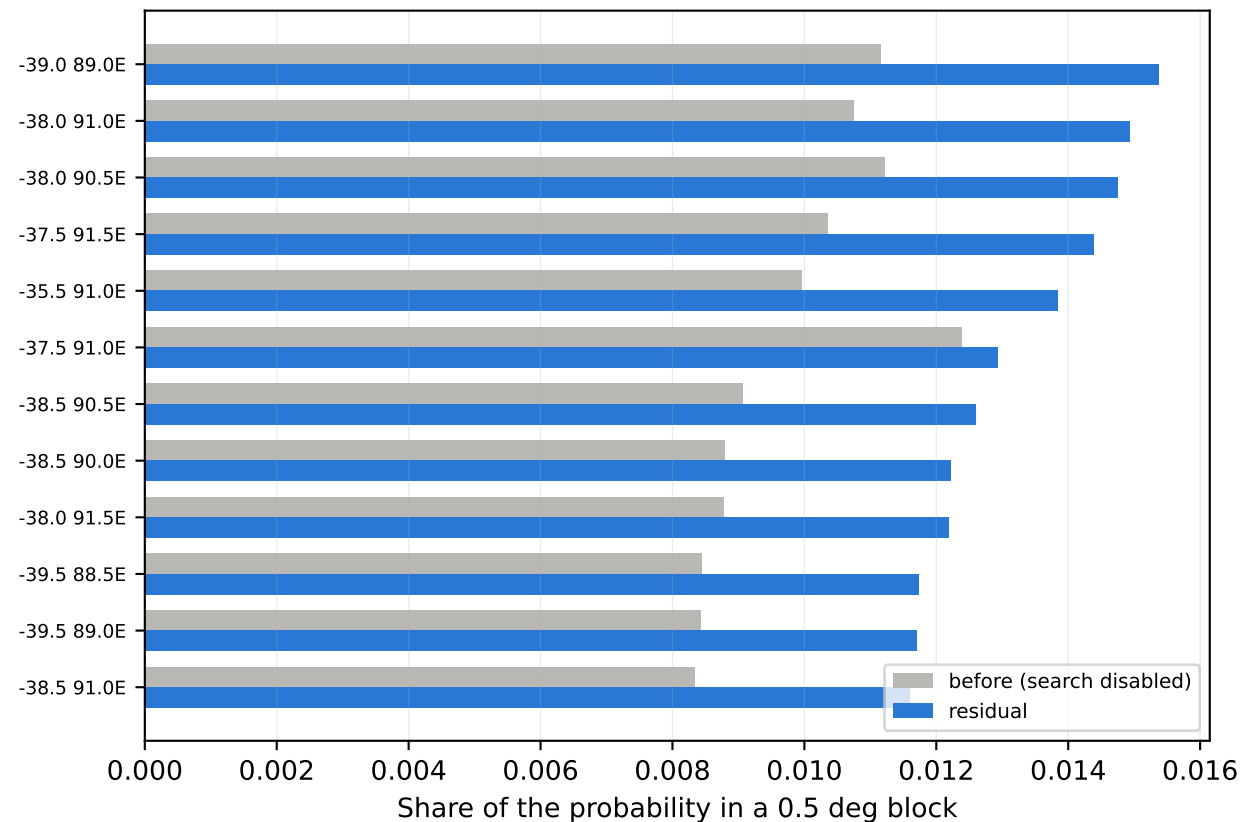




Davey eq. 11.2: probability of success, planning P_D 0.9



The best candidate areas, and what the search did to them



Seabed search evidence: numbers

IMPACTS FROM END-OF-FLIGHT MODULE END-OF-FLIGHT; 00:19 DATA OPTION NONE, LOG-ON CAUSE OTHER, CONSTRAINT +ALIVE (PROVISIONAL-OVERNIGHT)
Run eof-289-full: 4 replicates, 12,799,968 impacts; before: median -36.95, 95% -40.37 to -26.79; split-half 0.973; Kish ESS 11,042,066; on Phase 2 coverage 0.311

Mass removed at rho 0 (1 - evidence):
alone: bluefin-2014 0.0000
adding, in order: bluefin-2014 0.0000
alone: phase2-2014-2017 0.2941
adding, in order: phase2-2014-2017 0.2941
alone: ocean-infinity-2018 0.0363
adding, in order: ocean-infinity-2018 0.3301

scenario	Z	median	q025	q975	N of 33S	S of 39.5	on P2	ESS	split	vs before
run.toml	0.7206	-36.90	-40.61	-26.17	0.140	0.107	0.047	8,322,028	0.972	0.874
rho 0	0.7059	-36.89	-40.62	-26.13	0.143	0.109	0.027	8,015,062	0.971	0.864
rho 0.02	0.7118	-36.89	-40.61	-26.14	0.142	0.108	0.035	8,139,253	0.971	0.868
rho 0.1	0.7353	-36.90	-40.59	-26.20	0.137	0.105	0.066	8,616,404	0.972	0.883
rho 0.2	0.7647	-36.91	-40.56	-26.27	0.132	0.101	0.101	9,161,420	0.973	0.900
rho 0.3	0.7941	-36.92	-40.53	-26.34	0.127	0.097	0.134	9,640,362	0.973	0.916
rho 0.5	0.8529	-36.93	-40.48	-26.47	0.119	0.090	0.194	10,379,622	0.973	0.944
Phase 2 q 0.90	0.7339	-36.90	-40.59	-26.20	0.138	0.105	0.064	8,588,950	0.972	0.882
Phase 2 q 0.98	0.7103	-36.89	-40.62	-26.14	0.142	0.108	0.033	8,107,232	0.971	0.867
Phase 2 split, shared misses	0.7187	-36.89	-40.61	-26.16	0.141	0.107	0.044	8,282,232	0.972	0.873
Phase 2 split, independent misses	0.7168	-36.89	-40.61	-26.16	0.141	0.107	0.042	8,241,694	0.972	0.872
+ OI 2025-26 inferred, outboard band only	0.7165	-36.92	-40.61	-26.16	0.141	0.107	0.047	8,292,633	0.972	0.873
+ OI 2018 inferred, coverage 0.889	0.6864	-37.07	-40.64	-26.20	0.123	0.112	0.049	8,034,314	0.972	0.889
+ OI 2018 inferred, coverage 0.952	0.6840	-37.08	-40.64	-26.21	0.122	0.112	0.049	7,990,653	0.972	0.890

Residual PDF view. Source posterior: eof-289-full, 00:19 option none; search evidence applied with run.toml. Davey eq. 11.2 at a planning P_D of 0.9 on 0.5 deg blocks:

rank	candidate	area	residual mass	P(find)	km2	same block, search disabled
1	-39.00	89.00 E 0.50 deg	0.0154	0.0138	2411	0.0111
2	-38.00	91.00 E 0.50 deg	0.0149	0.0134	2444	0.0108
3	-38.00	90.50 E 0.50 deg	0.0148	0.0133	2444	0.0112
4	-37.50	91.50 E 0.50 deg	0.0144	0.0130	2461	0.0104
5	-35.50	91.00 E 0.50 deg	0.0138	0.0125	2524	0.0100
6	-37.50	91.00 E 0.50 deg	0.0129	0.0116	2461	0.0124
7	-38.50	90.50 E 0.50 deg	0.0126	0.0113	2427	0.0091
8	-38.50	90.00 E 0.50 deg	0.0122	0.0110	2427	0.0088

P(find) 25% needs the best 24 blocks, 59,005 km2
P(find) 50% needs the best 70 blocks, 171,921 km2
P(find) 75% needs the best 248 blocks, 623,095 km2